



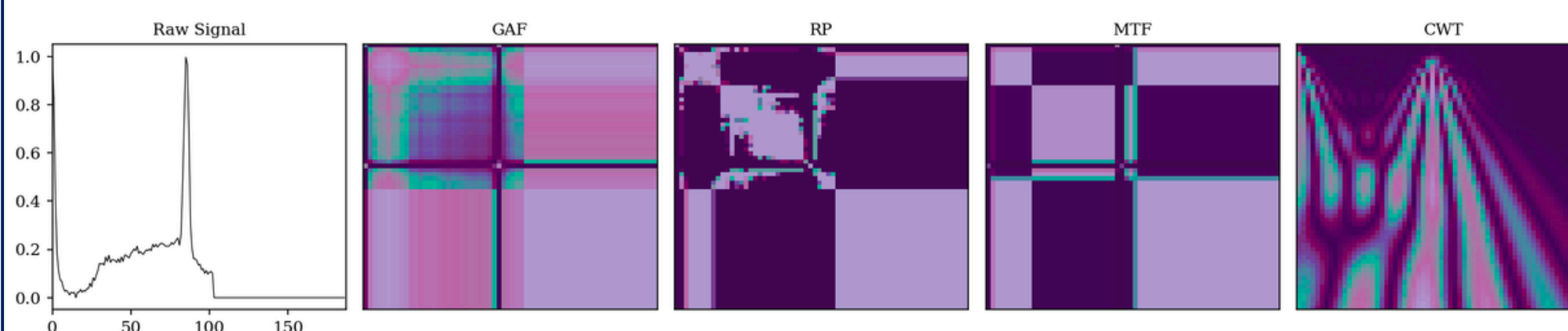
Spring School
AI For Impact

T-MECA: Trustworthy Multi-Encoding Classification of Arrhythmias

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Introduction & Objectives

Automated ECG classification achieves high accuracy, but existing methods lack uncertainty estimation and clinical referral, limiting real-world deployment.



Objectives

- Quantify per-class encoding importance via channel ablation
- Decompose uncertainty into model limitation vs data ambiguity
- Build a selective referral system based on prediction confidence

Main Contribution

- Encoding Importance:** Channel ablation shows CWT is the most critical encoding, with F1 drops up to 0.42 for pathological classes.
- Uncertainty Decomposition:** MC-Dropout separates model error from data noise: over 99% of uncertainty is inherent signal ambiguity.
- Clinical Decision Support:** An uncertainty-based referral system flags the 5% most uncertain cases, raising accuracy to 99.09%.

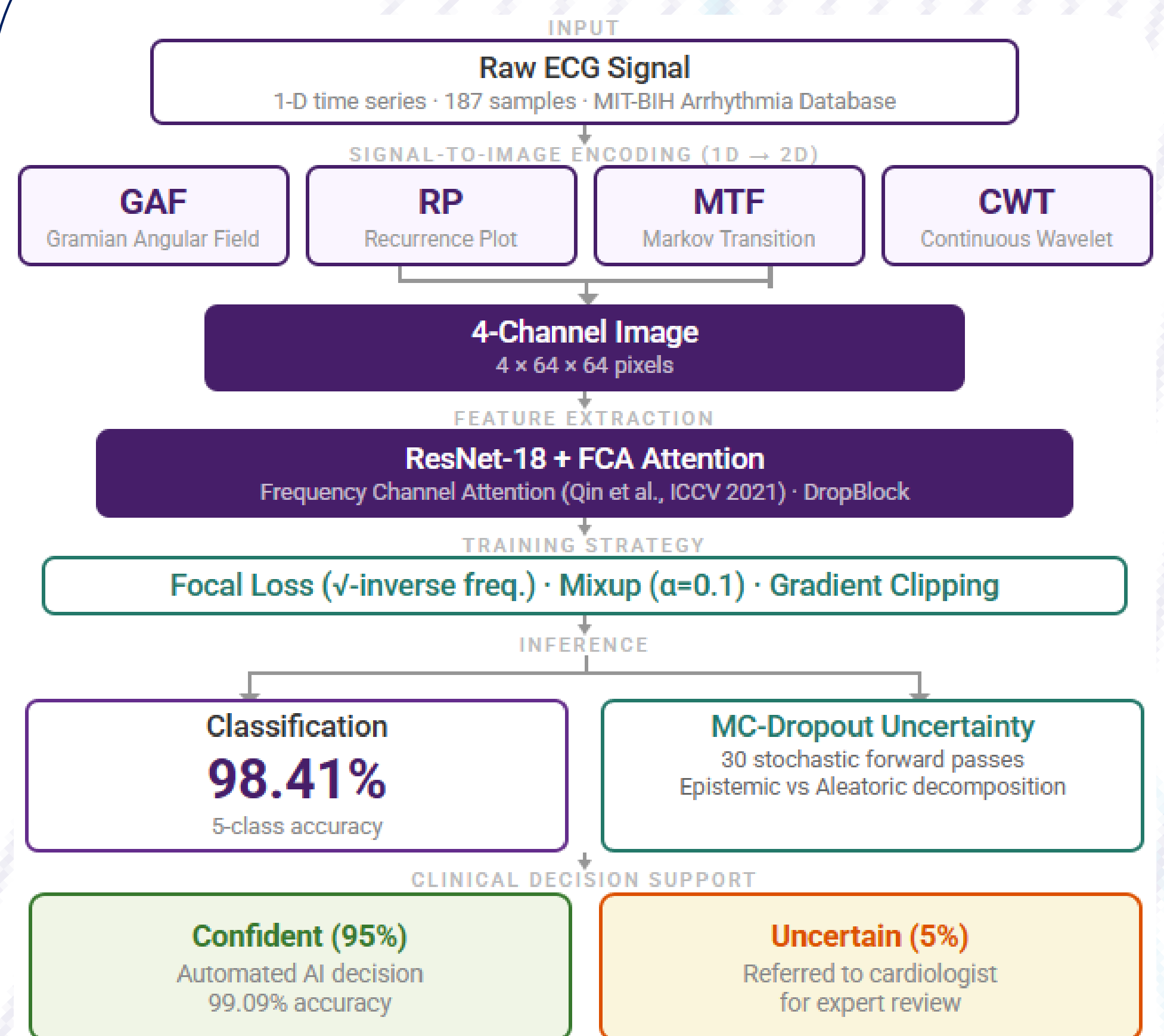
Conclusion & Perspectives

T-MECA achieves 98.41% accuracy with full uncertainty quantification. By referring only 5% of uncertain cases to a cardiologist, accuracy rises to 99.09%, bridging the gap between AI performance and clinical trust. These results open perspectives for uncertainty-guided active learning and cross-dataset validation on PTB-XL.

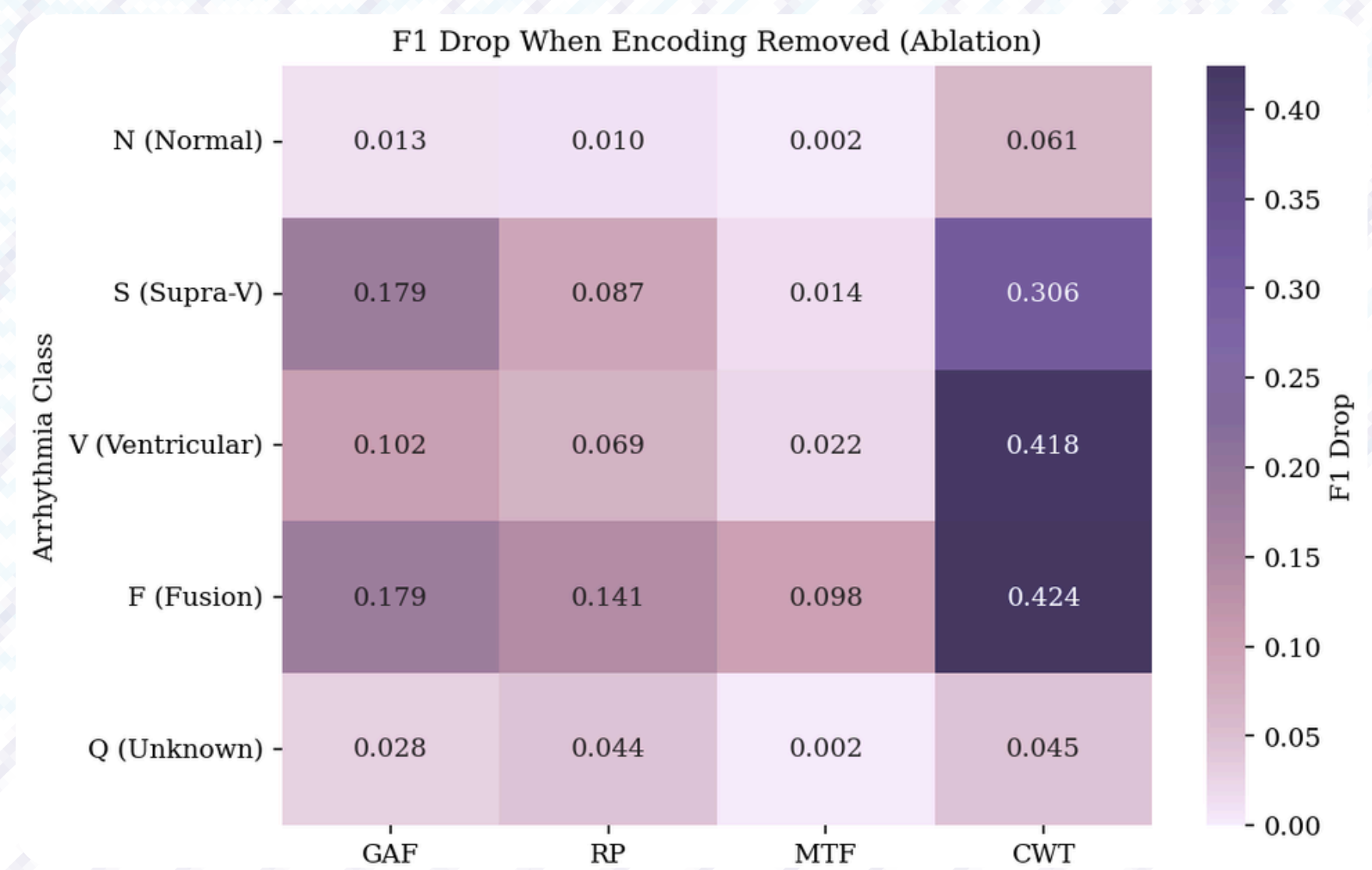
Bibliography

- [1] Moody & Mark, PhysioNet/CinC, 2001 [3] Gal & Ghahramani, MC Dropout, ICML 2016
[2] Qin et al., FcaNet, ICCV 2021 [4] Wang & Oates, Encoding Methods, 2015

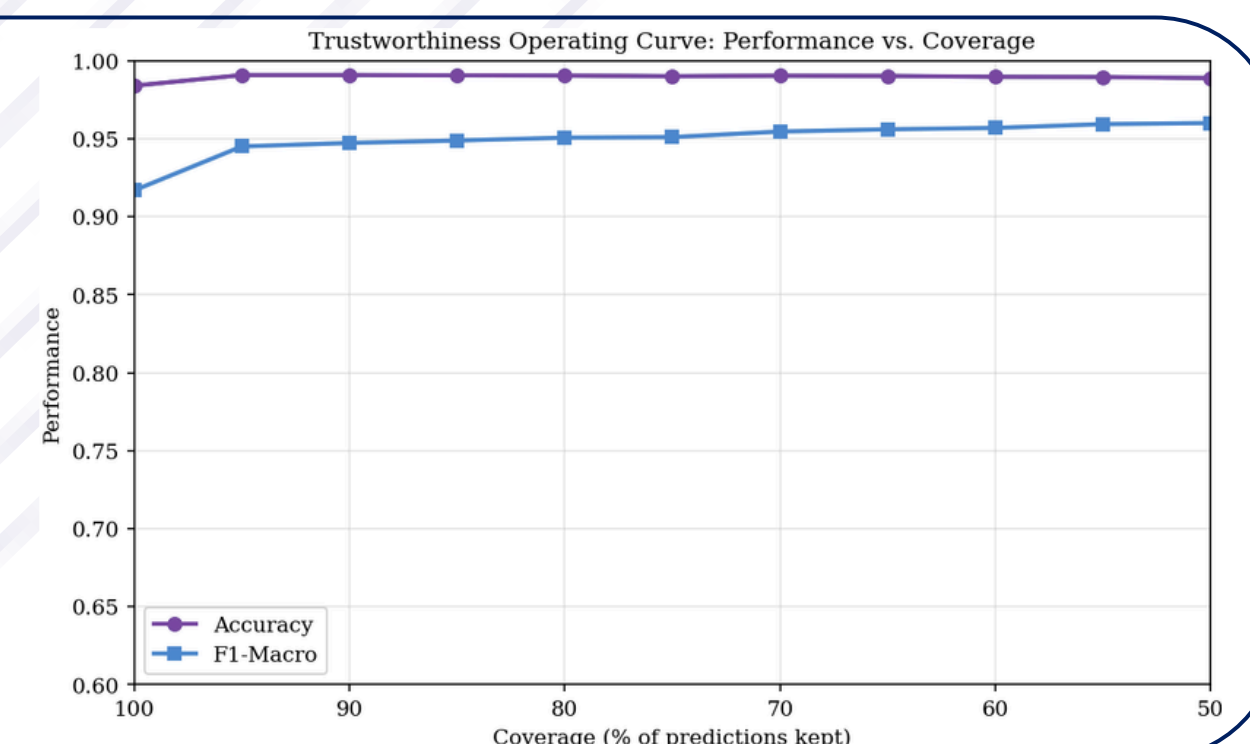
Methodology, Analysis & Results



98.41% ACCURACY **0.916** F1-MACRO **81.48%** F-CLASS PREC. **99.09%** CONFIDENT ACC



CWT is the most critical encoding: removing it drops F1 by up to 0.42 for pathological classes.



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